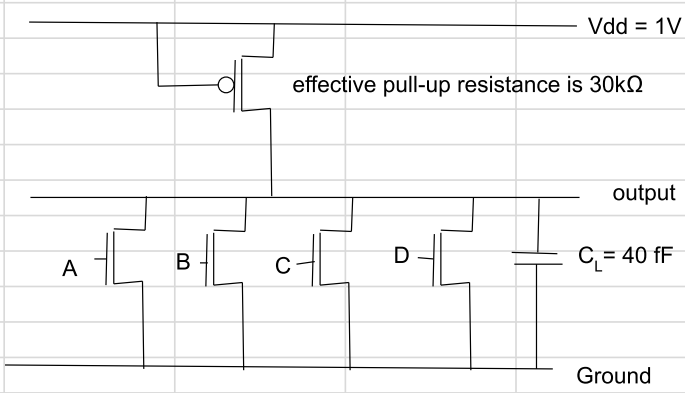
5A

Pseudo-NMOS NOR with one always-on PMOS pull-up load and four NMOS pull-down devices in parallel

I apologise for the trouble! Due to my disability, my wrists and hands are not working well at the moment so I must type all my work. I hope that I have formatted my work in a clear enough manner to mark, but if not, please let me know and I will explain my reasonings over a Zoom call or in person.



V ol = Voltage (output low)

Vol = Vdd(Rn/(Rp+Rn))

One NMOS pulldown on:

1*(10/(30+10))	equals Vol
0.25	**V**

4 NMOS pulldown on:

Resistance (total) = 10kOhms/4 because they are in parallel.

2.5	Ohms
1*(2.5/(30+2.5))	equals V ol
0.07692307692	**V**

Vol is input pattern dependent, because it depends on the resistance of the pull down network which is input-pattern dependent.

5B

NMOS ON	Voltage OL	R	
1	0.25	40	(30+10)
4	0.07692307692	32.5	(30+2.5)
	V	kOhms	

P=IV I=V/R ---> P=(V^2)/R

NMOS ON	V	R	P	**P**
1	1	40000	0.000025	**25**
4	1	32500	0.000030769230	**30.7**
		Ohms	Watts	**MicroWatts**

Falling output delay:

Pseudo NMOS does not output 0 ever. Shifts the falling output delay?

	Tau PHL =						
	300/40	7.5	Req in kOhms		ln ((vdd-vol)/(0.5-vol))		
	Vdd = 1	Vol = 0.25	---->	ln(0.75/0.25)			
	CMOS:						
	Tau PHL =	ln2* Rfalling *Cl					
	Req	10kOhms					
	Cl	40fF					
	Type	NMOS ON	Voltage OL	**Time**	**(for 1 NMOS on)**		
	Pseudo NMOS	1	0.25	**329.5836866**	**ps**		
5C	Static CMOS	1	0	**277.2588722**	**ps**		
	Rising output delay:		(no NMOS on)				
	Static cmos:	Worst case pull up resistance is 40kOhms					
	time =	Cl*resistance*ln2					
	time (ps) =	**1109.035489**	**for static cmos**				
	pseudo nmos:	Pull up resistance is 30kOhms					
	time =	Cl*resistance from 0.25V to 1V*(time factor)			ln ((vdd-vol)/(vdd-0.5))		
	time (ps) =	**486.5581297**	**for pseudo nmos**		equals Ln 1.5?		
	For a low power design, I would use the static CMOS design due to the pseudo-nmos' need for an always-on PMOS.						
5D	For a reduced transistor count, the pseudo NMOS design would be better, as there is only one PMOS transistor instead of 4 in series as required by the CMOS design.						
	For a reduced input capacitance, PMOS gates usually require more capacitance driving them. Reducing the number of PMOS gates is beneficial for this scenario, and so the pseudo nmos transistor would be preferred.						
5E							
	This would be bad due to the large string (6 of them!!) of PMOS transistors in series causing incredibly large resistance and delays. A better solution would be to separate the function using De Morgan, perhaps by splitting the inputs into two.						
	NOR ((NAND A,B,C),(NAND D,E,F)) maybe??						Ask yang wu